Monte Carlo Pricing of a European Call: Milstein Discretisation under Geometric Brownian Motion

Quant Finance Portfolio — Phase 2, Block 1, Algorithm 3

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Abstract

We specialise the Milstein scheme of the previous theoretical writeup to geometric Brownian motion, derive the resulting recursion, and apply it to Monte Carlo pricing of the European call. The principal purpose of the algorithm is not improvement of the pricer itself — Milstein and Euler share weak order 1 and produce statistically indistinguishable European prices — but rather *empirical demonstration of the strong/weak convergence asymmetry.* Running both schemes on the same grid of step sizes and on the same Brownian paths, we observe in a single experiment that Milstein achieves strong order 1 where Euler achieves only $\frac{1}{2}$, while the two are tied at weak order 1. This is the most direct possible confirmation of the theoretical result, and it is the experimental heart of Phase 2 Block 1.

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1 Introduction

The previous writeup of Block 1.2.1 implemented Euler-Maruyama and verified empirically that its strong order is $\frac{1}{2}$ and its weak order is 1. The asymmetry is a striking prediction of the SDE convergence theory developed in Block 1.2.0: the same scheme, applied to the same dynamics, has different convergence rates depending on which kind of error one measures.

The natural follow-up question is whether higher-order schemes lift both rates simultaneously, or only one. The answer, predicted by the theory, is that the Milstein scheme lifts the strong order from $\frac{1}{2}$ to 1 but leaves the weak order unchanged at 1. The present writeup verifies this prediction empirically on the same benchmark (European call pricing under GBM with all its inputs at canonical values) used in Block 1.2.1, with the additional design choice that the Euler and Milstein experiments share their step-size grid so that the two slopes can be plotted on a single log-log axis.

The pricing application of the algorithm is, frankly, a sideshow. Both schemes give weak order 1 for European pricing under GBM, and the closed-form sampler of Block 1.1 is unbiased and faster than either; Milstein adds nothing to the pricing pipeline that Euler did not already provide. The value of the algorithm is in the convergence study, where the asymmetry is demonstrated in code rather than asserted by a theorem. The practical relevance of Milstein in this project will arise in Block 4 (pathwise sensitivity), where the strong order matters because the simulated path must be a faithful pathwise approximation of the true process for path-derivative estimators to make sense.

2 Milstein for geometric Brownian motion

2.1 The recursion

Recall the GBM SDE under the risk-neutral measure $\mathbb{Q}$:

$$\mathrm{d}S_t \;=\; rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t, \qquad S_0 > 0, \tag{1}$$

so $a(s) = rs$ and $b(s) = \sigma s$, hence $b'(s) = \sigma$ and $b(s)b'(s) = \sigma^2 s$. The Milstein recursion of the previous writeup specialises to

$$\hat{S}_{n+1} \;=\; \hat{S}_n + r\hat{S}_n h + \sigma\hat{S}_n\Delta W_n + \tfrac{1}{2}\sigma^2\hat{S}_n\big[(\Delta W_n)^2 - h\big], \qquad \hat{S}_0 = S_0, \tag{2}$$

with $\Delta W_n \sim \mathcal{N}(0,h)$ independent across n and $h = T/N$. Factoring $\hat{S}_n$ out:

$$\hat{S}_{n+1} \;=\; \hat{S}_n \cdot \Big(1 + rh + \sigma\Delta W_n + \tfrac{1}{2}\sigma^2\big[(\Delta W_n)^2 - h\big]\Big). \tag{3}$$

The recursion remains multiplicative, but the multiplicative factor is now *quadratic* in ΔW_n rather than linear as in Euler. This has two practical consequences worth noting.

2.2 Improved positivity

The Euler factor $1 + rh + \sigma\Delta W_n$ is unbounded below along the negative direction: a sufficiently negative ΔW_n produces a negative factor and hence a negative $\hat{S}_{n+1}$. The Milstein factor adds the term $\frac{1}{2}\sigma^2((\Delta W_n)^2 - h)$, which has *no upper bound on the negative side*: the $(\Delta W_n)^2$ component is always nonnegative, and only the $-\sigma^2 h/2$ piece pulls the factor down, by an amount bounded by $\sigma^2 h/2$. Concretely, the Milstein factor is bounded below by

$$1 + rh + \sigma\Delta W_n - \tfrac{1}{2}\sigma^2 h, \tag{4}$$

which is the worst case where $(\Delta W_n)^2 = 0$. For typical finance parameters this lower bound is essentially the Euler factor, so the positivity behaviour of Milstein is no worse than that of Euler. Empirically, the Milstein-$\sigma^2/2$ correction more than compensates the loss of the $\sigma\Delta W_n$ term in most realisations: the factor $1 + rh + \sigma\Delta W_n + \frac{1}{2}\sigma^2((\Delta W_n)^2 - h)$ is, on average, closer to 1 than the Euler factor.

We do not implement positivity defences in the algorithm; for the parameter regime of our experiments, neither Euler nor Milstein produces nonpositive prices in any realisation we have observed.

2.3 The corrector term and its expectation

The new term $\frac{1}{2}\sigma^2((\Delta W_n)^2 - h)$ is the Milstein corrector. Two of its features deserve special note.

It has zero mean. Since $(\Delta W_n)^2$ has mean $\mathrm{Var}(\Delta W_n) = h$, the corrector has expectation zero: $\mathbb{E}[\frac{1}{2}\sigma^2((\Delta W_n)^2 - h)] = 0$. This is the mathematical reason that Milstein and Euler tie at weak order 1: the corrector affects each path differently, but its average contribution to $\mathbb{E}[f(\hat{S}_T)]$ is, to leading order, zero.

It is pathwise nonzero. Although the corrector vanishes on average, on any individual path $(\Delta W_n)^2 - h$ takes a specific value, typically of order h in absolute terms. This is the term that, in Euler, was discarded and contributed $O(h)$ to the strong error per step. In Milstein, it is included, and the next omitted term is of higher order: the strong error becomes $O(h)$ globally instead of $O(\sqrt{h})$.

The two features together explain the asymmetry. Strong convergence sees the pathwise contribution of the corrector and benefits from its inclusion. Weak convergence sees only the average contribution, which is zero either way. This is the conceptual content of the strong/weak distinction made concrete on a specific recursion.

2.4 Convergence orders

By Theorem 6.1 of the previous writeup applied to GBM,

$$\mathbb{E}[|S_T - \hat{S}_T^{\text{Milstein}}|] \;=\; O(h), \qquad |\mathbb{E}[f(S_T)] - \mathbb{E}[f(\hat{S}_T^{\text{Milstein}})]| \;=\; O(h) \quad \text{for } f \text{ smooth enough.} \tag{5}$$

Compare with Euler:

$$\mathbb{E}[|S_T - \hat{S}_T^{\text{Euler}}|] \;=\; O(h^{1/2}), \qquad |\mathbb{E}[f(S_T)] - \mathbb{E}[f(\hat{S}_T^{\text{Euler}})]| \;=\; O(h). \tag{6}$$

Strong order: $\frac{1}{2}$ (Euler) vs 1 (Milstein) — a genuine factor-$\sqrt{h}$ improvement at small h. Weak order: 1 (Euler) vs 1 (Milstein) — no improvement.

3 The biased estimator and its comparison with Euler

3.1 Definition

Let $\hat{S}_T^{(i),\,\text{Milstein}}$ for $i = 1, \ldots, M$ denote independent realisations of the Milstein-discretised terminal price. The Monte Carlo Milstein estimator of the European call price is

$$\hat{C}_{M,h}^{\text{Milstein}} \;\stackrel{\text{def}}{=}\; \frac{1}{M}\sum_{i=1}^{M} e^{-rT}\,(\hat{S}_T^{(i),\,\text{Milstein}} - K)^+. \tag{7}$$

The bias-variance decomposition of Block 1.2.1, equation (8), applies here verbatim: total mean square error decomposes as variance/M plus bias-squared, with bias $O(h)$.

3.2 Comparison with Euler at fixed budget

The pricing performance of Milstein and Euler is, in this setting, indistinguishable. Both have weak order 1, so both produce a bias of order h for the same h. The constants in front of the h asymptote may differ slightly between the two, but for canonical GBM parameters the difference is too small to matter at the sample sizes where the bias is detectable.

The cost per path of Milstein is marginally higher: one extra multiplication and one squaring per Euler step (the corrector $(\Delta W_n)^2 - h$). The overhead is on the order of 20–30% in our implementation, well below the cost of generating the ΔW_n themselves. So at fixed total budget, Milstein delivers roughly the same accuracy as Euler with marginally fewer paths or marginally larger h. Either way the difference is in the noise.

The genuine practical disadvantage of Milstein for European pricing, visible in the validation suite of Section 4, is none whatsoever for European pricing as such: this whole comparison is academic. The advantage materialises in pathwise applications. We mention it here because confronting the irrelevance of Milstein for vanilla European pricing is part of internalising the strong/weak distinction.

4 Empirical validation strategy

The validation is the experimentally most informative part of the algorithm. We design a single comparison study that exhibits both the strong-order improvement and the weak-order tie of Milstein relative to Euler.

4.1 Test 1: side-by-side strong order

The most distinctive part of the validation script is a comparison of the two schemes' strong errors on the *same* grid of step counts and using the *same* Brownian paths. The protocol:

1. Choose a logarithmic grid of step counts N_k matching the strong-order grid of Block 1.2.1, $N \in \{10, 30, 100, 300, 1000, 3000\}$.
2. For each N_k, sample M matrices of Brownian increments ΔW of shape (M, N_k).
3. Run the Euler scheme on these increments to obtain $\hat{S}_T^{\text{Euler},(i)}$ and the Milstein scheme on the *same* increments to obtain $\hat{S}_T^{\text{Milstein},(i)}$.
4. Compute the exact terminal $S_T^{(i)}$ from $W_T^{(i)} = \sum_n \Delta W_n^{(i)}$ via the closed-form GBM solution.
5. Estimate $\text{strong}_{\text{scheme}}(h_k) \approx M^{-1} \sum_i |S_T^{(i)} - \hat{S}_T^{\text{scheme},(i)}|$ for both schemes.
6. Plot both error curves on a single log-log axis.

The expected result is two parallel lines on log-log: Euler with slope $\frac{1}{2}$, Milstein with slope 1. Beyond confirming the two convergence orders, the side-by-side plot makes the asymmetry visually obvious: at the smallest h tested, Milstein's strong error should be roughly $\sqrt{h_{\text{smallest}}/h_{\text{largest}}}$ times smaller than Euler's, an order of magnitude or more.

4.2 Test 2: weak order under common random numbers

A direct adaptation of the Block 1.2.1 weak test, applied to Milstein. The expected slope is 1, the same as Euler. This test serves as the empirical confirmation that Milstein *does not* improve the European pricer: same weak order, same asymptotic quality.

The grid restriction discussed in the Block 1.2.1 numerical lesson applies here: the bias must dominate the residual MC noise of the CRN-corrected difference. With $M = 200{,}000$ and $N \in \{10, 20, 50, 100\}$, both schemes resolve the slope cleanly.

4.3 Tests 3 and 4: coherence and validation

Identical in structure to the corresponding tests of Block 1.2.1, applied to the Milstein pricer. No conceptual novelty.

5 Algorithmic specification

Algorithm 1 Milstein Monte Carlo for the European call (GBM)

Require: Spot S_0, strike K, maturity T, rate r, volatility σ, step count N, sample size M, confidence level $1-\beta$, random generator `rng`.

Ensure: Estimate $\hat{C}$, half-width h_M, sample variance S_M^2.

```
h ← T/N,    D ← e^{-rT}
for i = 1, ..., M do
    S^(i) ← S_0
    for n = 0, 1, ..., N − 1 do
        Draw ΔW ~ N(0, h) using rng
        S^(i) ← S^(i) · (1 + rh + σΔW + (1/2)σ^2((ΔW)^2 − h))
    end for
    Y_i ← D · max(S^(i) − K, 0)
end for
Ĉ ← M^{-1} Σ_i Y_i
S_M^2 ← (M − 1)^{-1} Σ_i (Y_i − Ĉ)^2
h_M ← z_{β/2} √(S_M^2 / M)
return (Ĉ, h_M, S_M^2)
```

The implementation vectorises the inner loop across paths (advance all M states simultaneously) using `numpy.cumprod` on the matrix of factors, identical in structure to the Block 1.2.1 Euler implementation. The only changes to the code are the addition of the quadratic corrector term in the factor and the corresponding docstrings; the Block 1.2.2 implementation is barely 30 lines longer than Block 1.2.1.

6 Bibliographic notes

The Milstein scheme is from Milstein (1974) and is treated in detail in Kloeden & Platen [3], Section 10.3. The empirical side-by-side comparison study designed here is a direct adaptation of the comparison study in Higham [2], Section 5. For the practical relevance of Milstein in finance applications, Glasserman [1, Section 6.3] provides the context-specific discussion: when does the strong order of a scheme matter for the financial applications at hand.

References

[1] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.

[2] D. J. Higham, *An algorithmic introduction to numerical simulation of stochastic differential equations*, SIAM Review, 43(3):525–546, 2001.

[3] P. E. Kloeden and E. Platen, *Numerical Solution of Stochastic Differential Equations*, Springer, 1992.